# Final Report

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## Abstract

This project addresses the challenges clients and loan providers face in connecting efficiently in the loan marketplace, particularly underserved groups such as young borrowers and new immigrants. We propose a centralized platform that balances the needs of clients and loan providers by incorporating an interactive matching system. By enabling both parties to match with each other based on their preference for each loan or offer posts they make, the platform fosters financial inclusion and promotes local economic growth.

## Intro

This project aims to address the gap in the current loan borrowing landscape by creating a platform where both clients and loan providers can efficiently connect with each other based on their specific needs.  It is challenging for clients to identify and connect with the most appropriate loan providers for their specific financial needs, while loan providers struggle to find potential borrowers in need of their services.

From the client's perspective, many individuals, especially those with non-traditional backgrounds (e.g., young borrowers, new immigrants, or individuals without a solid credit history), struggle to find lenders willing to work with them. This creates barriers for building a credit score and achieving financial stability. On the other hand, from the lending business perspective, loan providers, especially smaller businesses, face challenges in identifying and connecting with potential borrowers, limiting their ability to grow, gain trust with the community, and generate profit.

The lack of platforms that cater equitably to both parties has created an imbalance in the marketplace. Existing solutions often prioritize borrower-driven searches, leaving loan providers with limited tools to proactively seek clients. This project proposes a system designed to bridge this gap, ensuring an interactive and balanced experience for both borrowers and lenders.

## Solution System

To address these challenges, we developed a matching system with two primary user roles: **clients** and **loan providers**.

- **Clients** create loan request posts that specify attributes such as:
    - Loan type
    - Term length
    - Maximum acceptable interest rate
    - Desired loan amount
- **Loan providers** create loan offer posts defining attributes such as:
    - Loan type
    - Term length
    - Minimum interest rate

    - Minimum and maximum loan amount

The system evaluates compatibility between loan requests and offers by calculating a **match score**. Posts with a score above 60 are presented to both parties, enabling them to review, accept, or decline the match. Emails will be revealed to the other party if both accept the loan for communication.

This matching system streamlines the process of finding compatible loan opportunities, bridging the gap between clients and providers, and supporting greater inclusivity in the loan marketplace.

## Project Design

### Client User (Looking for a Loan)

A Client user is someone who requires financial assistance and is seeking a loan to meet specific needs. These needs could range from financial, real estate, business, or other needs. The client's role in the system is to create posts that clearly describe their loan request, outlining the amount, purpose, and repayment terms they are seeking.

**Client Registration:**

- A client first needs to register to Loan Matcher using their personal financial information in order to let Loan Provider users see what kind of loan the client is seeking to acquire.

**Client Login:**

- The client then can login using the account they just registered in order to gain access to all available features.

**Loan Request Details:** The client then creates a post with detailed information about the loan they require. This may include:

- **Loan Amount:** The specific amount of money the client is seeking.
- **Type/Category**: The reason for the loan e.g. business, real estate, financial, other.
- **Repayment Terms:** The client may suggest their preferred repayment period, interest rate, or flexibility in repayment.
- **Description:** Any other details that might help a loan provider assess the request, such as a client's credit history or financial standing.

**Posting Process:** After entering these details, the client can submit the post to the system. This post becomes visible to loan providers who can browse and offer their loans in response.

**Post Interaction:**

- **View Posts**: Clients will browse through post offers made by loan providers.

- **View own post:** Clients can view their own post they created.
- **Connect:** If a client is interested in a loan offer, they can accept and either wait until the loan provider also accepts the client's post, or immediately be able to view contact information if the loan provider accepted first.

**Loan Provider User (Offering a Loan)**
A Loan Provider user is an individual or institution offering loans to clients. Loan providers may have a set of standard loans or may be flexible, tailoring offers based on the client's specific needs.
**Loan Offer Creation:**

- **Login/Registration:** Like clients, loan providers must register or log in to offer loans and manage their posts.
- **Loan Offer Details:** Loan providers create posts that detail the loan offers they are providing. These posts include:
    - **Title:** To indicate to clients that they are offering a loan
    - **Loan Amount:** The range or exact amount of money available for lending.
    - **Interest Rates:** The rate of interest they charge on the loan.
    - **Repayment Terms:** The length of time over which the loan must be repaid.
    - **Type/Category**: The reason for the loan e.g. business, real estate, financial, other.
    - **Description:** Additional information provided such as specifying the criteria for clients to qualify for the loan (e.g., credit score, income level) or flexible repayment options.
- **Posting Process:** Once the loan details are filled out, the provider posts the offer on the system, making it visible to potential clients.

**Post Interaction:**

- **Browse Client Posts:** Loan providers interested in offering loans who they themselves have posted will accept the client's post, detailing the terms and conditions of the loan they are offering. Clients will be to see the loan provider's post of offer.
- **View own Offer:** Loan providers can view their own loan offer post
- **Connect:** If a loan provider finds a post that aligns with their loan terms, they can accept the post and either wait for the client to also accept, or be given contact information if the client has already accepted.

**Complete Transaction Flow**
A complete transaction cycle in the system can be visualized as follows:

1. Client Posts Loan Request: A client creates and posts a detailed loan request, specifying the amount, type, repayment terms, and a description.

2. Loan Posts Offer: A loan provider creates and posts a detailed loan offer, specifying the amount, rate, repayment terms, and a description.

3. Client Browses Offers: Clients review available posts and search for potential loan providers whose needs align with their request.

4. Loan Provider Browses Requests: Loan providers review available posts and search for potential clients whose needs align with their loan offerings.

5. Client Responds: After reviewing a loan provider's post, the client can either decline or accept the loan provider's post.

6. Loan Provider Responds: After reviewing a client's post, the loan provider can either decline or accept the client's post.

7. Mutual Acceptance: Once the client and loan provider finds each other's post they are comfortable with, they initiate a connection to move forward with the loan agreement.

8. Finalization: The client and loan provider finalize the loan terms, and both parties agree to proceed with the loan. The client can then receive the loan amount, and repayment begins according to the agreed-upon terms.

**Conclusion**

This system facilitates a dynamic and efficient marketplace for loan requests and offerings, where both clients and loan providers can interact, negotiate, and finalize agreements. By ensuring that each post includes relevant details and facilitating communication between the parties, the system supports a transparent and organized loan process, empowering both client users and loan provider users to achieve their financial goals.

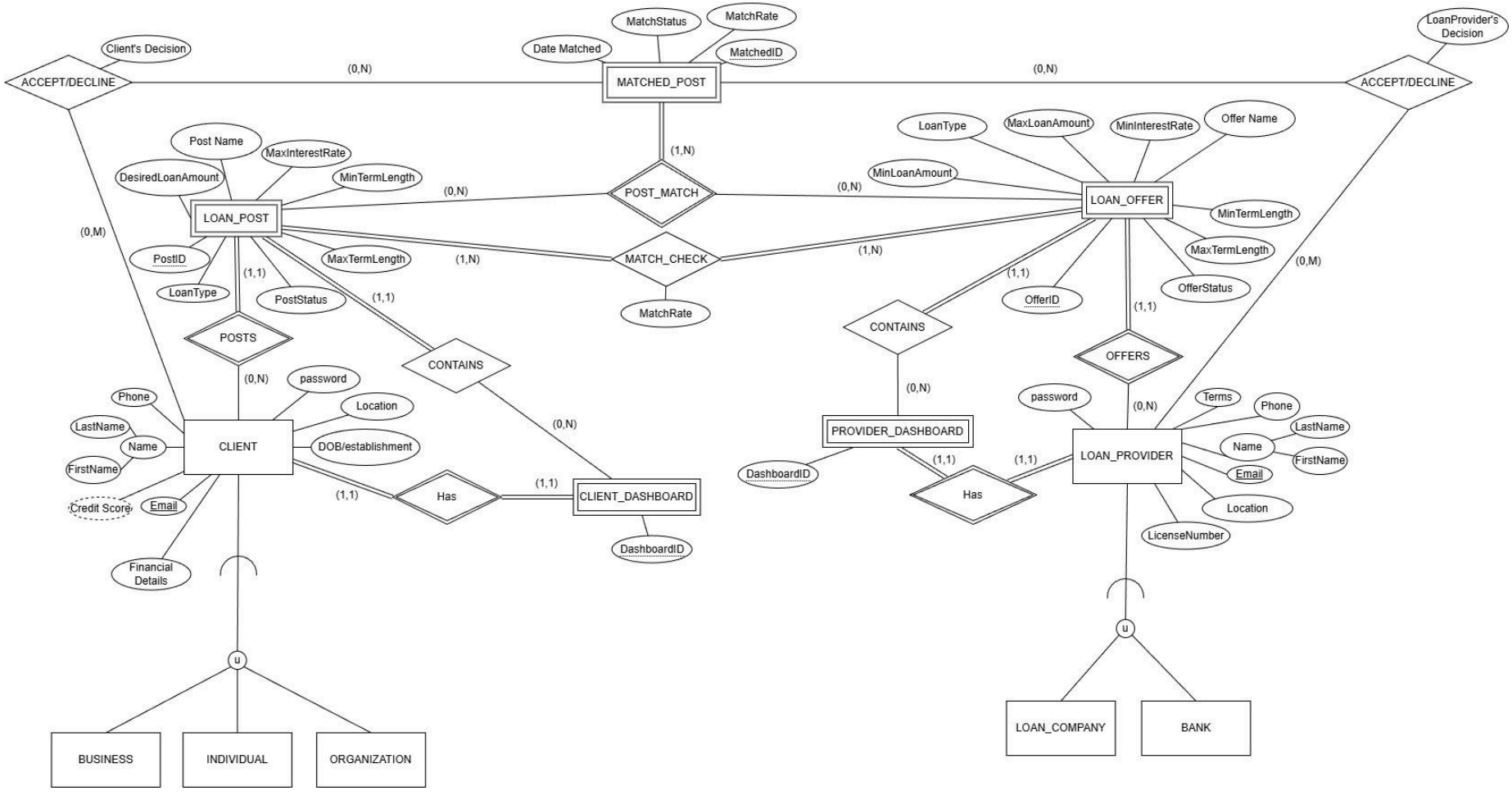
## **Project Implementation**- ER Diagram:

### Assumptions:

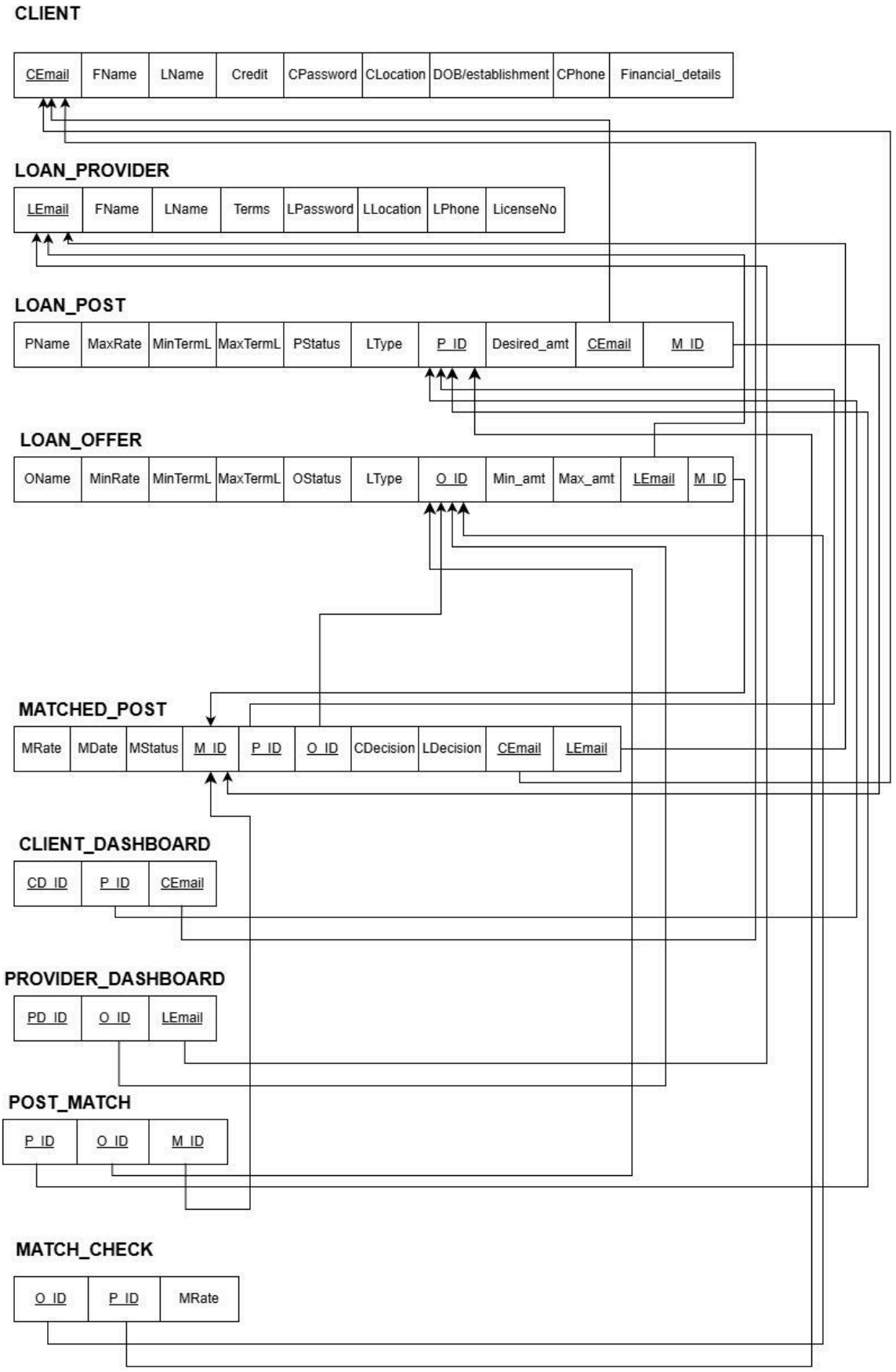
- Clients and loan providers are considered users
- If client was deleted, then loan post would be deleted as well
- Dashboard must contain the already created loan offer/post
- Loan post/offer must be contained in dashboard
- Admin authenticates loan providers and clients, and also has the abitlity to remove them
- The system matches loan posts and loan offers and creates a matched_post when the two can match
- The matched_post contains both the loan post and the loan offer matched
- Both the client and the post provider has the ability to accept/decline the post
- Once the matched post status is set the post/offer status will change
- Admin must authenticate user

### Changes:

- Removed Admin

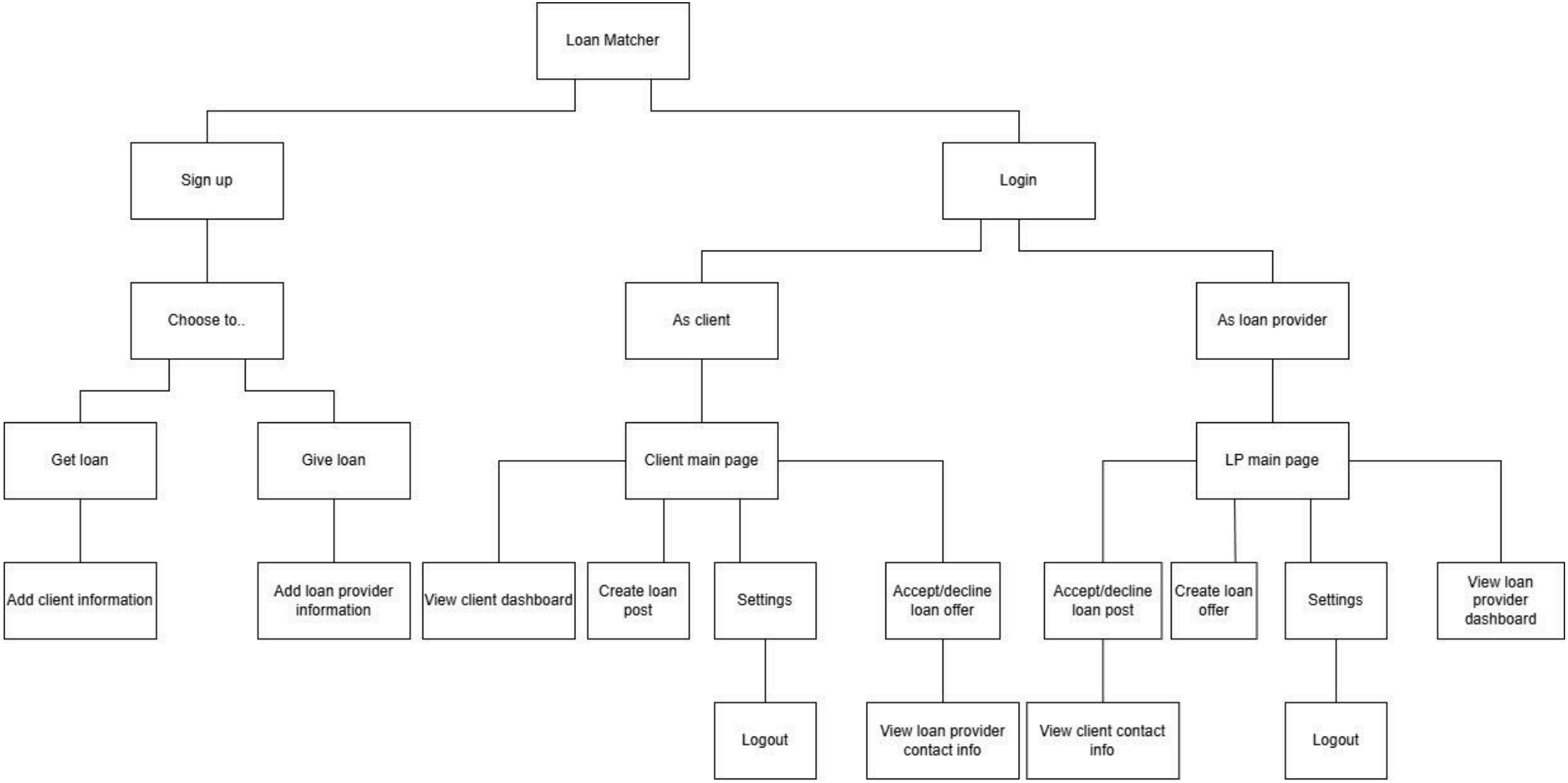


Relational Model Diagram:

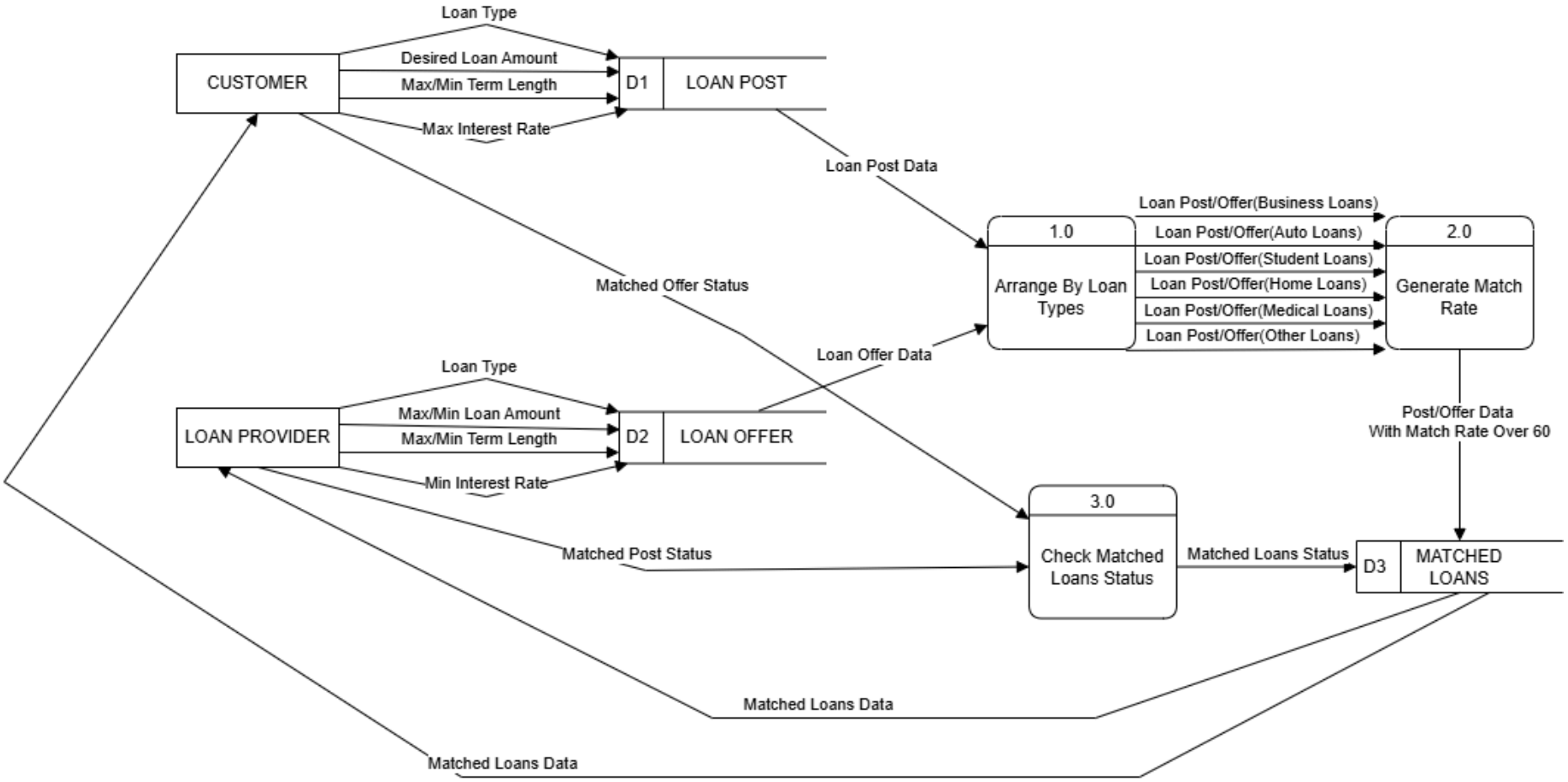



Changes made: Removed Admin

HIPO Diagram:



DFD Diagram:


Loan Type
CUSTOMER
Desired Loan Amount
Max/Min Term Length
D1
LOAN POST
Max Interest Rate
Loan Post Data
Loan Post/Offer(Business Loans)
1.0
Loan Post/Offer(Auto Loans)
2.0
Loan Post/Offer(Student Loans)
Arrange By Loan Types
Loan Post/Offer(Home Loans)
Generate Match Rate
Loan Post/Offer(Medical Loans)
Loan Post/Offer(Other Loans)
Matched Offer Status
Loan Offer Data
Loan Type
Post/Offer Data
With Match Rate Over 60
LOAN PROVIDER
Max/Min Loan Amount
Max/Min Term Length
D2
LOAN OFFER
Min Interest Rate
3.0
Matched Post Status
Check Matched Loans Status
Matched Loans Status
D3
MATCHED LOANS
Matched Loans Data
Matched Loans Data

**Project Implementation DSMS:**

For this project, we chose to use MySQL. MySQL is a database management system that leverages **Structured Query Language (SQL)** as its primary interface for interacting with the database. SQL is a standardized language used to manage and manipulate relational databases. MySQL implements SQL commands and extends its functionality with additional features, allowing users to efficiently retrieve, insert, update, and delete data.

SQL statements:

Signup client:

```
-- Check if the email already exists in the client table
SELECT CEmail FROM client WHERE CEmail = ?;

-- Check if the email already exists in the loan_provider table
SELECT LEmail FROM loan_provider WHERE LEmail = ?;

-- Insert the new client if email does not exist in any table
INSERT INTO client (CEmail, Fname, Lname, CPhone, CLocation,
DOB_Establishment, Credit, CPassword)
VALUES (?, ?, ?, ?, ?, ?, ?, ?);
```

Signup loan provider:

```
-- Check if the email already exists in the client table
SELECT CEmail FROM client WHERE CEmail = ?;

-- Check if the email already exists in the loan_provider table
SELECT LEmail FROM loan_provider WHERE LEmail = ?;

-- Insert the new client if email does not exist in any table
INSERT INTO loan_provider (LEmail, Fname, Lname, LPhone, LLocation, LicenseNo,
LPassword)
VALUES (?, ?, ?, ?, ?, ?, ?);
```

Login client:

```
-- Check if the email and password match
SELECT * FROM client WHERE CEmail = ? AND CPassword = ?;
```

Login loan provider:

```
-- Check if the email and password match
SELECT * FROM loan_provider WHERE LEmail = ? AND LPassword = ?;
```

CreateLoanPost:

```
-- Insert a new loan post into the loan_post table
```

```
INSERT INTO loan_post
(PTitle, Description, MaxRate, MinTermStart, MinTermEnd,
MaxTermStart, MaxTermEnd, PStatus, LType, Desired_amt, CEmail)
VALUES
(?, ?, ?, ?, ?, ?, ?, 'Pending', ?, ?, ?);
```

CreateLoanOffer:

```
-- Insert a new loan offer into the loan_offer table
INSERT INTO loan_offer
(OName, Description, MinRate, MinTermStart, MinTermEnd,
MaxTermStart, MaxTermEnd, OStatus, LType, Min_amt, Max_amt,
LEmail)
VALUES
(?, ?, ?, ?, ?, ?, ?, 'Pending', ?, ?, ?, ?);
```

ViewClientDashboard:

```
-- Get all loan posts for the client with complete information
   SELECT
       -- Loan Post Information
       LP.P_ID,
       LP.PName,
       LP.MaxRate,
       LP.MinTermL,
       LP.MaxTermL,
       LP.PStatus,
       LP.LType,
       LP.Desired_amt,

       -- Client Information
       C.CEmail,
       C.Fname,
       C.Lname,
       C.Credit,
       C.CLocation,
       C.Financial_details,
       C.CPhone,

       -- Match Information (if exists)
       MP.MRate AS Matched_Rate,
       MP.MDate AS Match_Date,
       MP.MStatus AS Match_Status,
       MP.CDecision AS Client_Decision,
```

```
        MP.LDecision AS Provider_Decision,

        -- Loan Provider Information (if matched)
        LP2.Fname AS Provider_Fname,
        LP2.Lname AS Provider_Lname,
        LP2.Terms AS Provider_Terms,
        LP2.LLocation AS Provider_Location,
        LP2.LPhone AS Provider_Phone,
        LP2.LicenseNo AS Provider_License

    FROM LOAN_POST LP
    JOIN CLIENT C ON LP.CEmail = C.CEmail
    LEFT JOIN MATCHED_POST MP ON LP.P_ID = MP.P_ID
    LEFT JOIN LOAN_PROVIDER LP2 ON MP.LEmail = LP2.LEmail
    WHERE LP.CEmail = p_CEmail
    ORDER BY
        CASE
            WHEN LP.PStatus = 'Active' THEN 1
            WHEN LP.PStatus = 'Matched' THEN 2
            WHEN LP.PStatus = 'Closed' THEN 3
            ELSE 4
        END,
        LP.MinTermL DESC;

    -- Get simple post count statistics
    SELECT
        COUNT(*) AS Total_Posts,
        COUNT(CASE WHEN PStatus = 'Active' THEN 1 END) AS Active_Posts,
        COUNT(CASE WHEN PStatus = 'Matched' THEN 1 END) AS Matched_Posts,
        COUNT(CASE WHEN PStatus = 'Closed' THEN 1 END) AS Closed_Posts
    FROM LOAN_POST
    WHERE CEmail = p_CEmail;
```

ViewLoanProviderDashboard:

```
-- Get all loan offers for the provider with complete information
    SELECT
        -- Loan Offer Information
        LO.O_ID,
        LO.OName,
        LO.MinRate,
        LO.MinTermL,
        LO.MaxTermL,
        LO.OStatus,
        LO.LType,
        LO.Min_amt,
```

```
    LO.Max_amt,

    -- Loan Provider Information
    LP.LEmail,
    LP.Fname,
    LP.Lname,
    LP.Terms,
    LP.LLocation,
    LP.LPhone,
    LP.LicenseNo,

    -- Match Information (if exists)
    MP.MRate AS Matched_Rate,
    MP.MDate AS Match_Date,
    MP.MStatus AS Match_Status,
    MP.CDecision AS Client_Decision,
    MP.LDecision AS Provider_Decision,

    -- Client Information (if matched)
    C.Fname AS Client_Fname,
    C.Lname AS Client_Lname,
    C.Credit AS Client_Credit,
    C.CLocation AS Client_Location,
    C.Financial_details AS Client_Financial_Details,
    C.CPhone AS Client_Phone

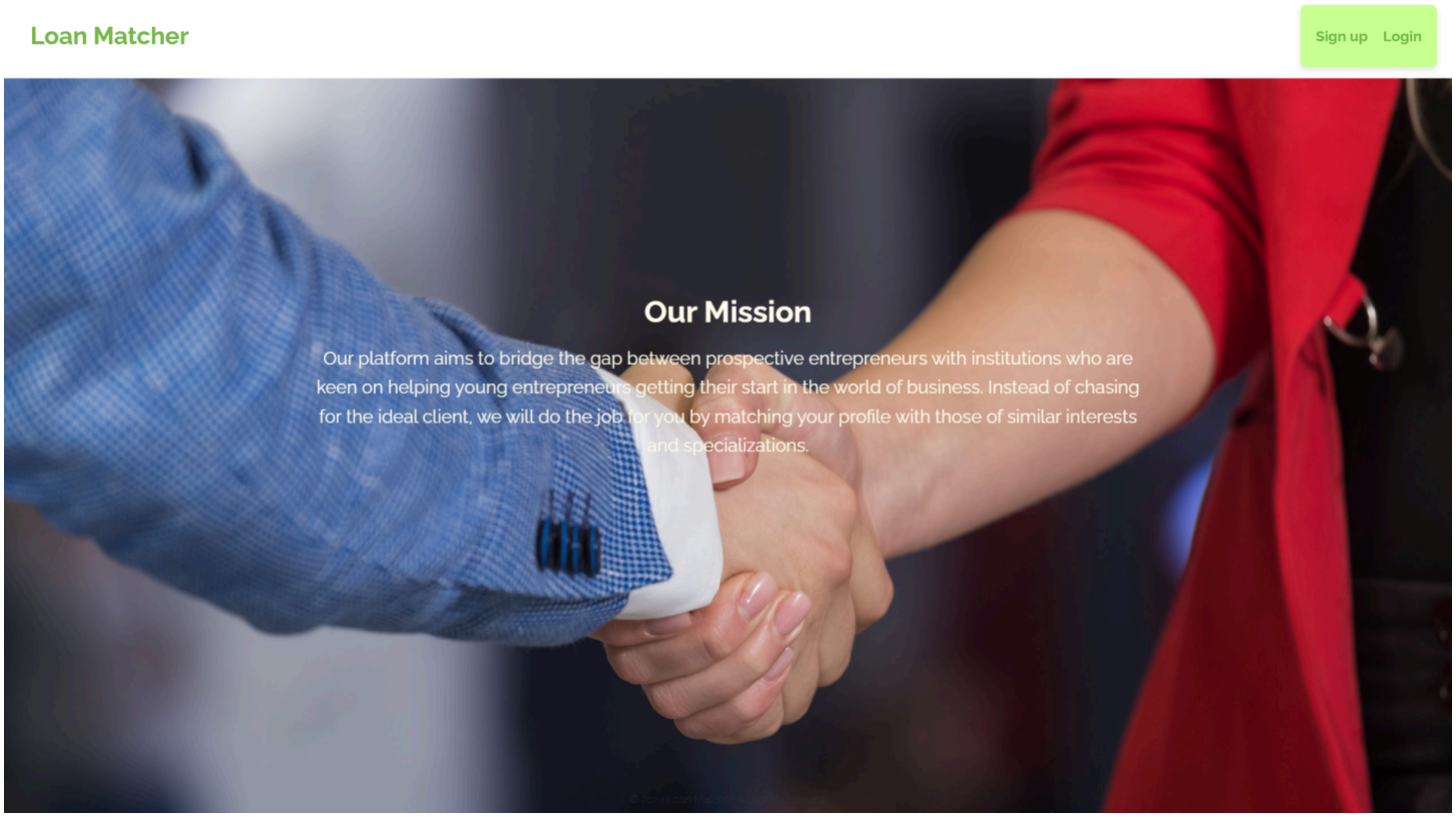
FROM LOAN_OFFER LO
JOIN LOAN_PROVIDER LP ON LO.LEmail = LP.LEmail
LEFT JOIN MATCHED_POST MP ON LO.O_ID = MP.O_ID
LEFT JOIN CLIENT C ON MP.CEmail = C.CEmail
WHERE LO.LEmail = p_LEmail
ORDER BY
    CASE
        WHEN LO.OStatus = 'Active' THEN 1
        WHEN LO.OStatus = 'Matched' THEN 2
        WHEN LO.OStatus = 'Closed' THEN 3
        ELSE 4
    END,
    LO.MinTermL DESC;

-- Get simple offer count statistics
SELECT
    COUNT(*) AS Total_Offers,
    COUNT(CASE WHEN OStatus = 'Active' THEN 1 END) AS Active_Offers,
    COUNT(CASE WHEN OStatus = 'Matched' THEN 1 END) AS Matched_Offers,
    COUNT(CASE WHEN OStatus = 'Closed' THEN 1 END) AS Closed_Offers
```

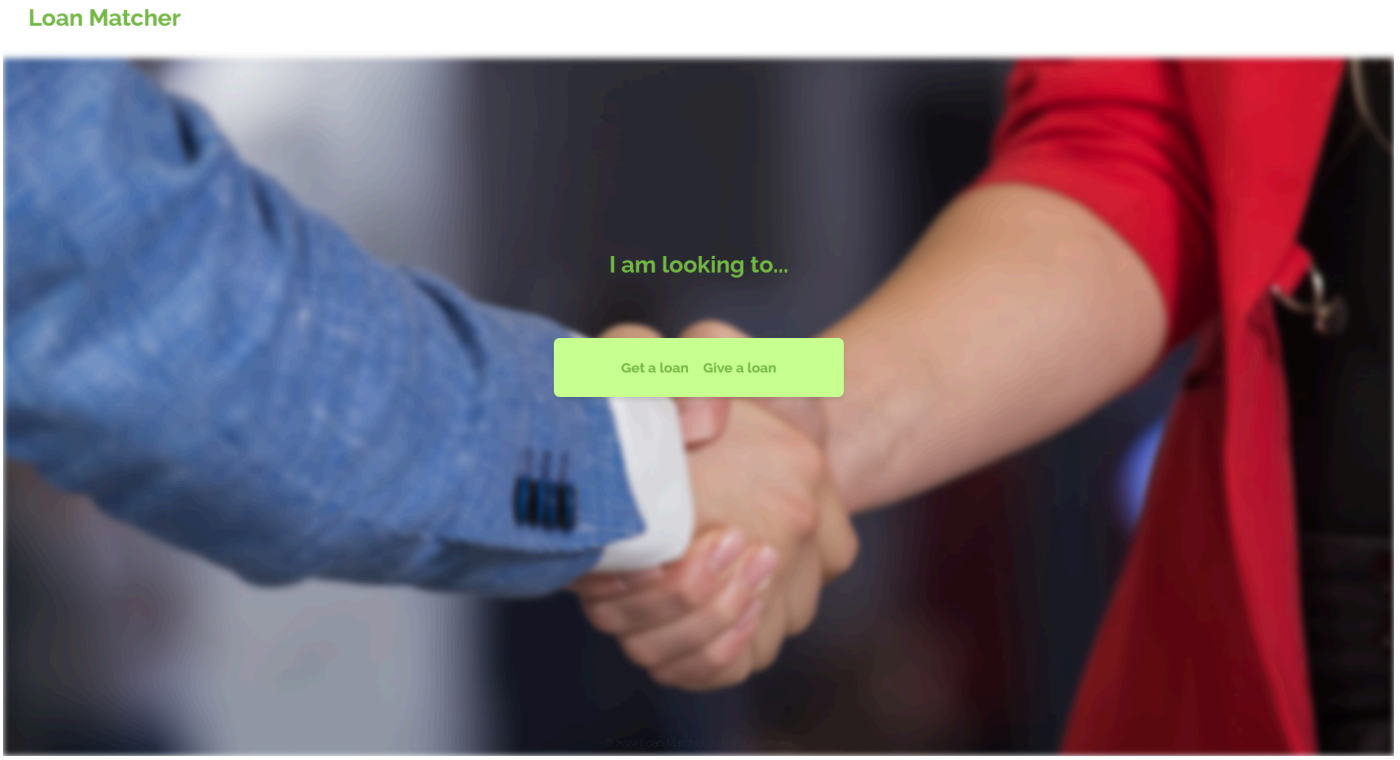
```
    FROM LOAN_OFFER
    WHERE LEmail = p_LEmail;
```

**User Manual:**

Landing page:

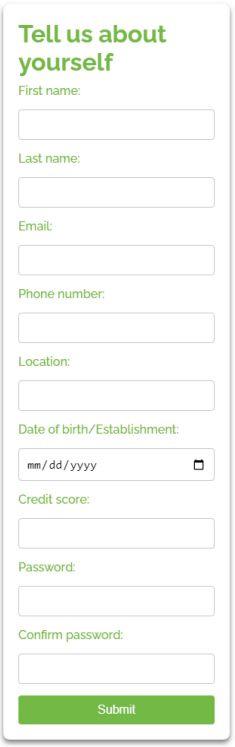



Signup page:



Client signup page:

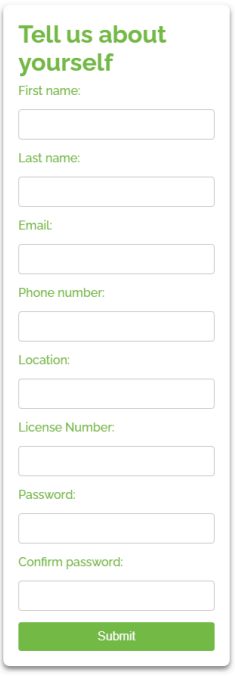
Loan Matcher




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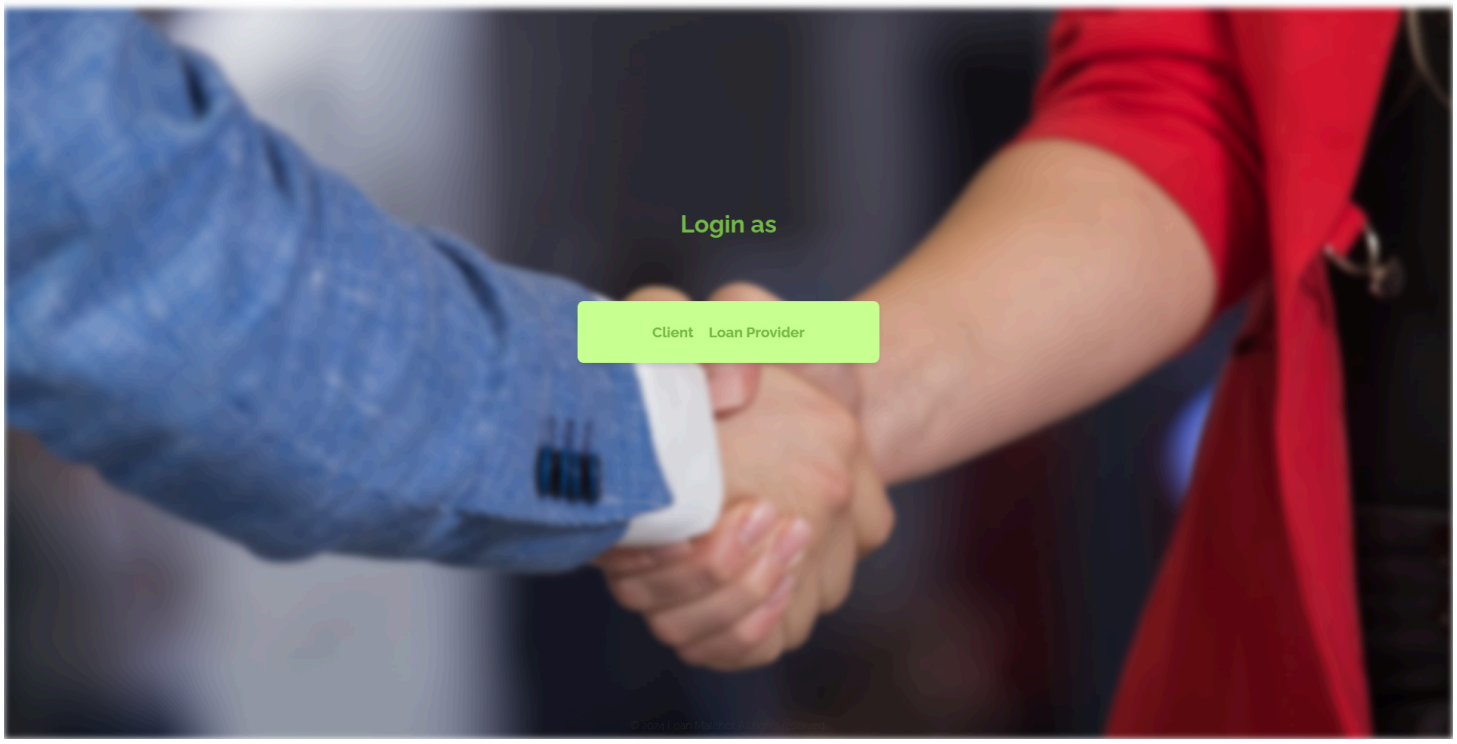
Loan Provider signup page:

Loan Matcher

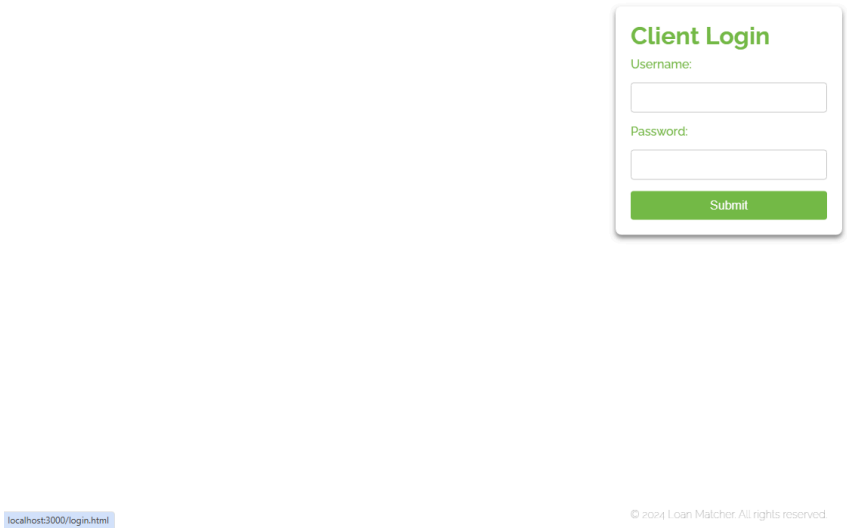



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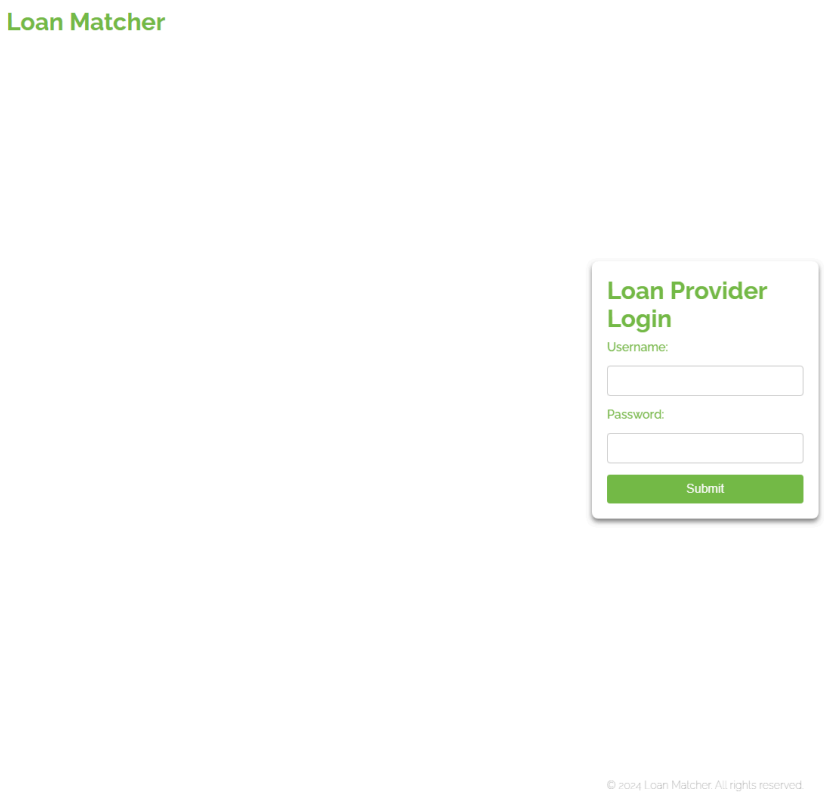
Login page:



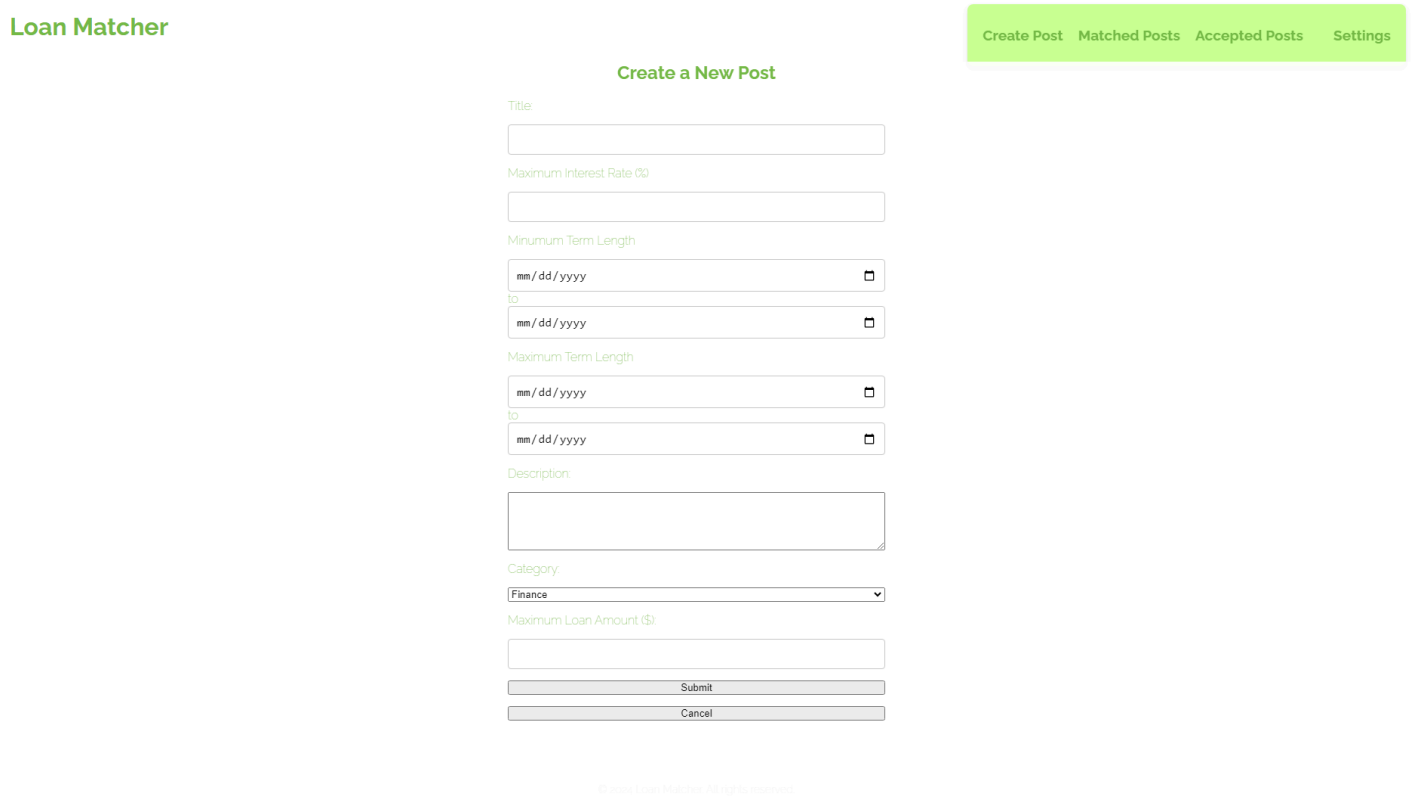

Client login page:



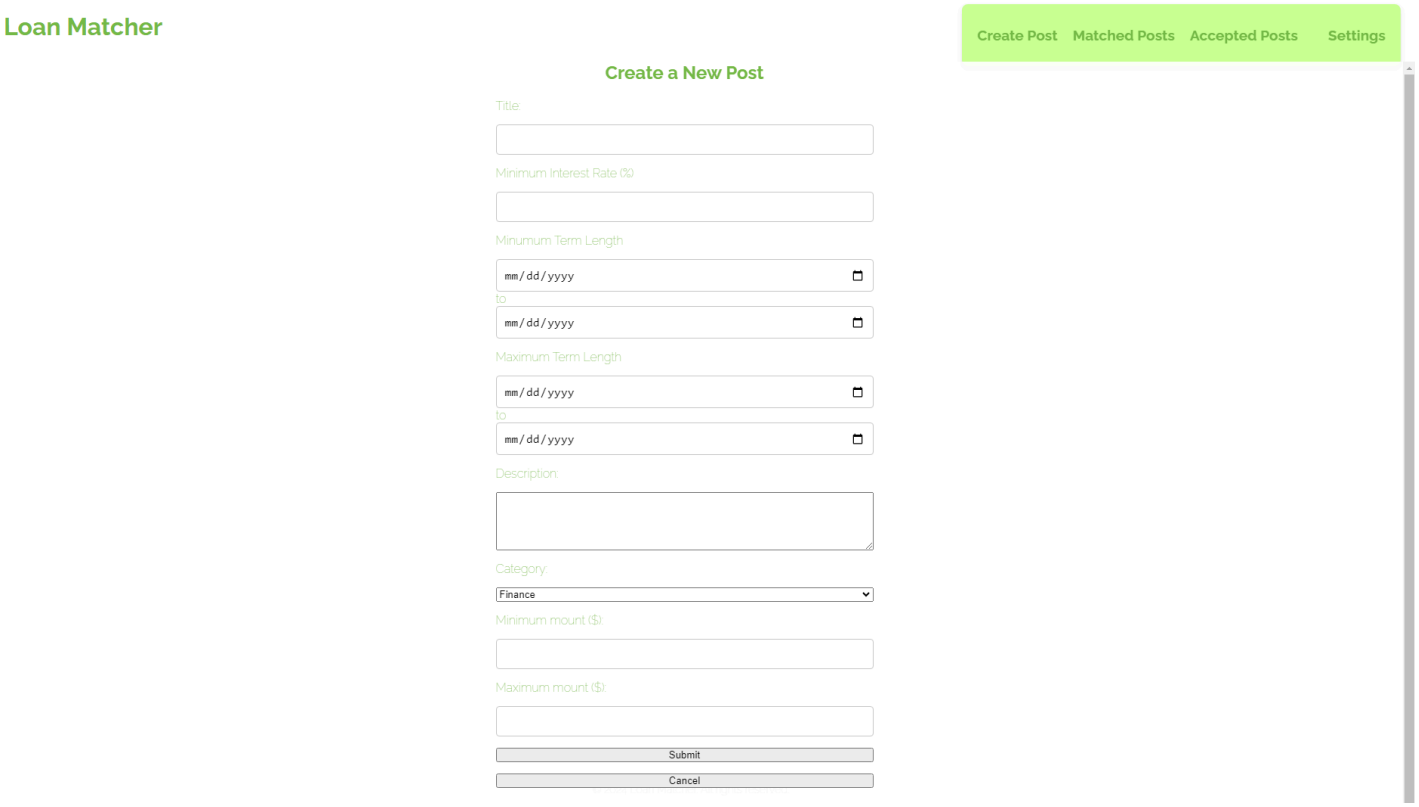
Loan Provider login page:



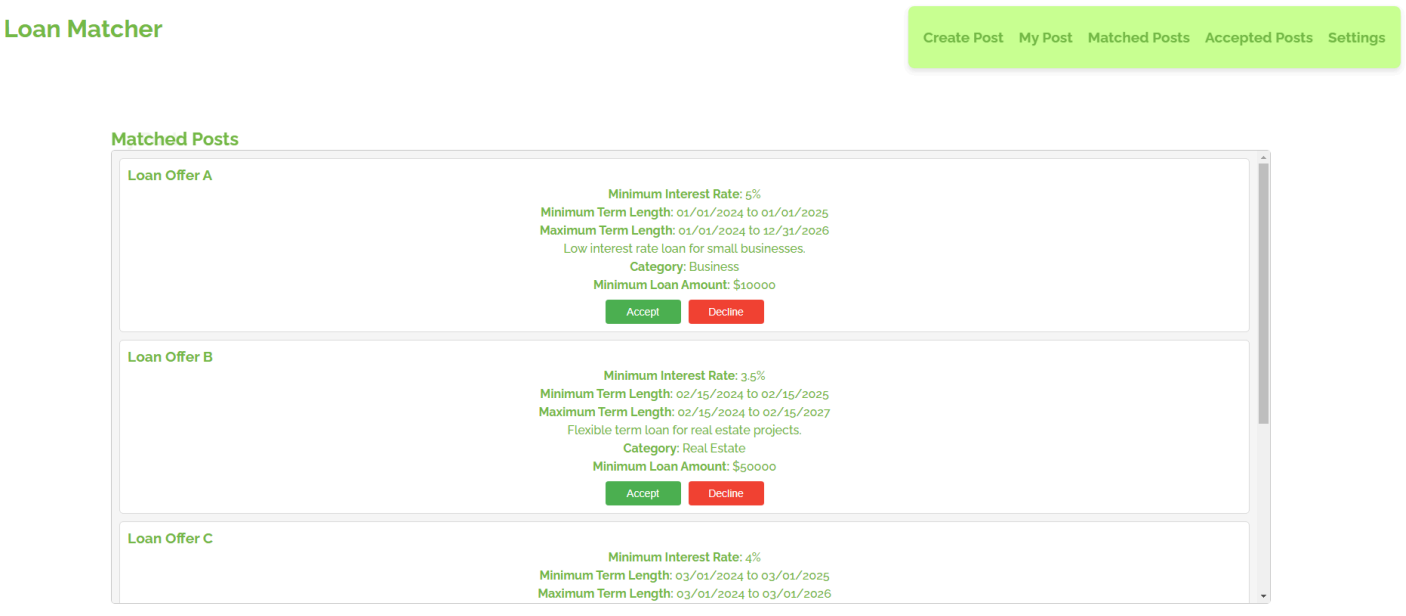

Client main page “Create Post” tab:



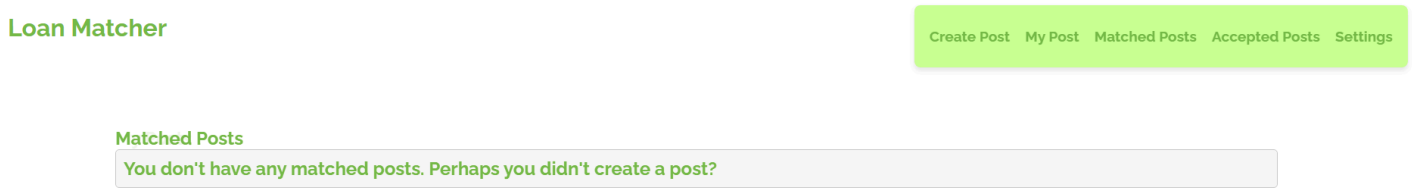
LP main page “Create Post” tab:




Client/LP main page “Matched Posts” tab populated:



Client/LP main page “Matched Posts” tab empty:

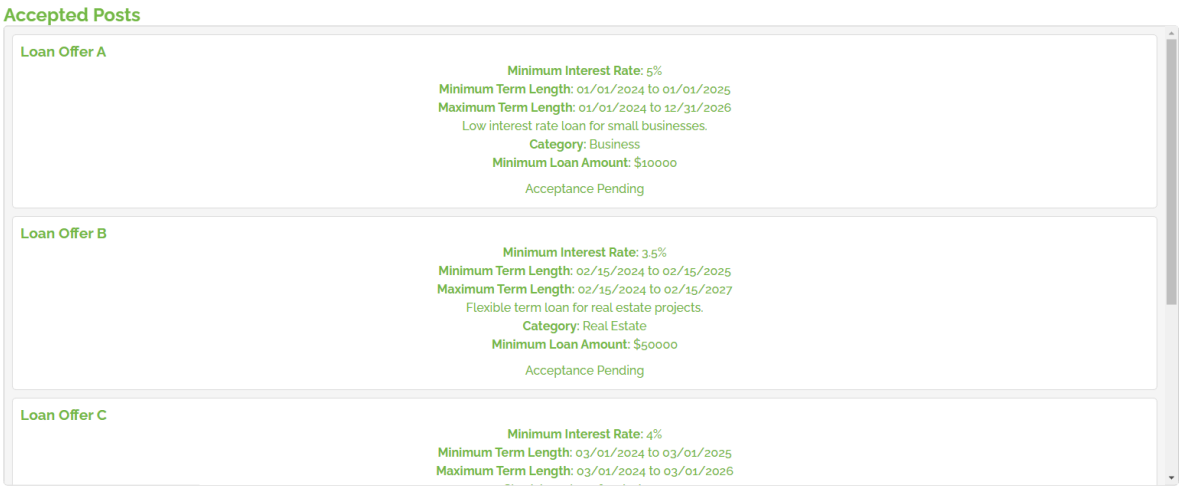




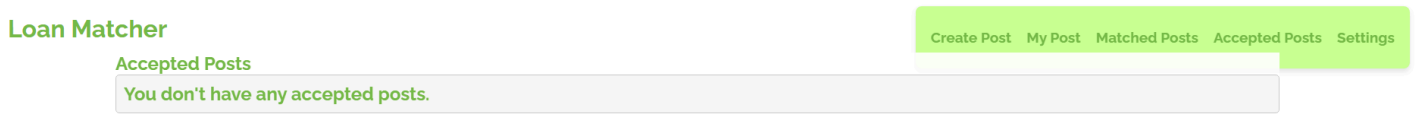


Client/LP main page “Accepted Posts” tab populated:

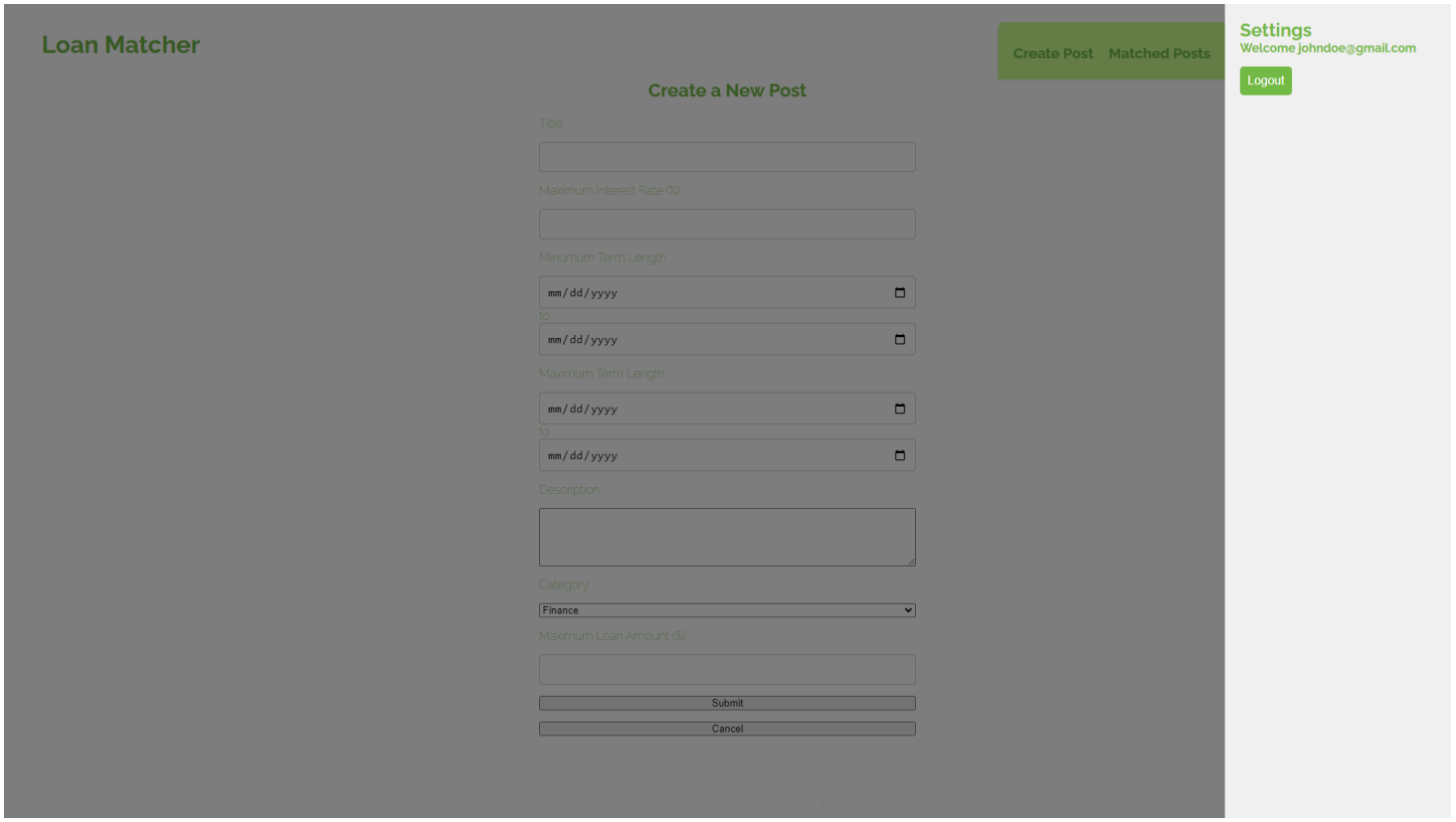


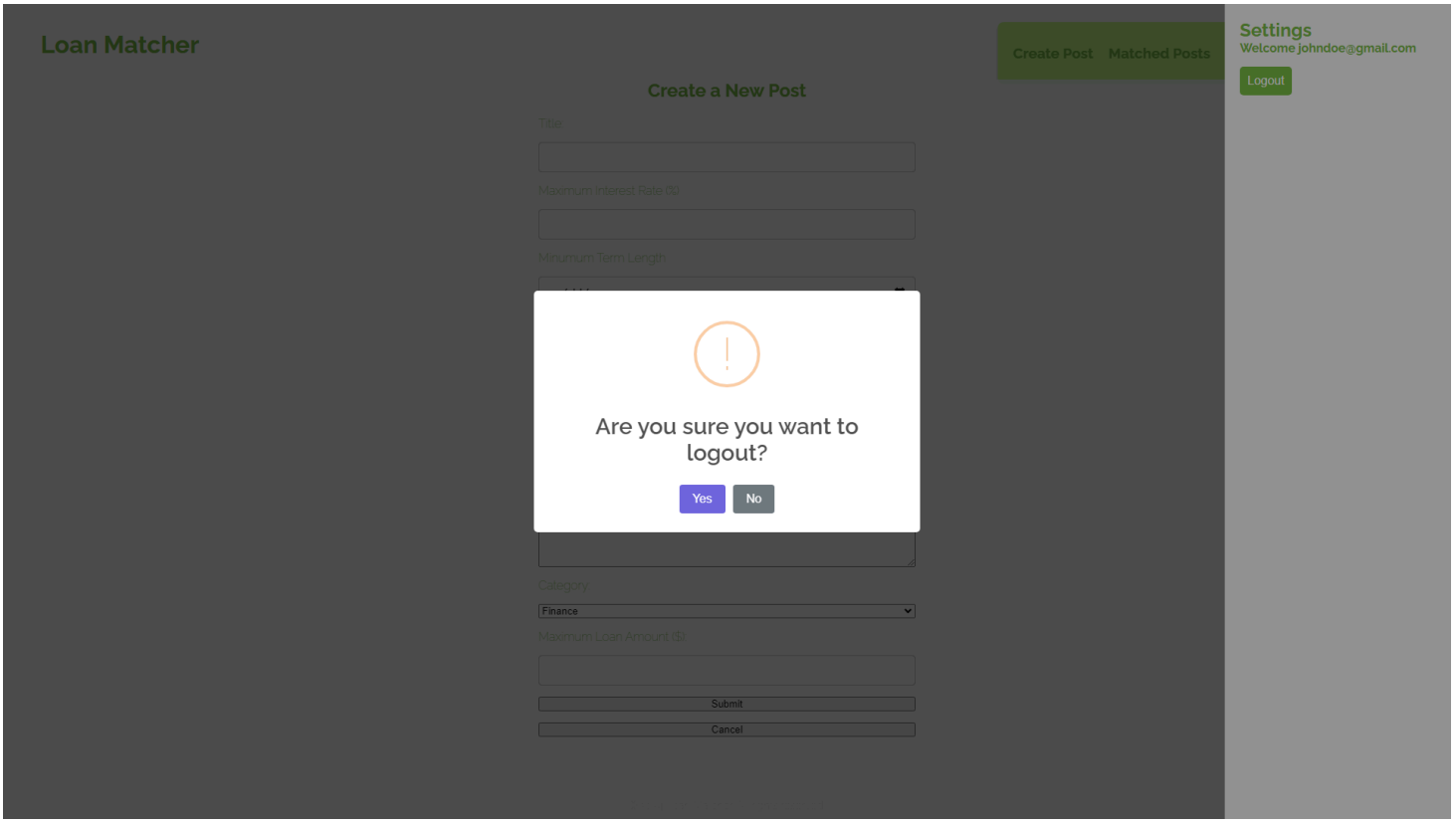
Client/LP main page “Accepted Posts” tab empty:



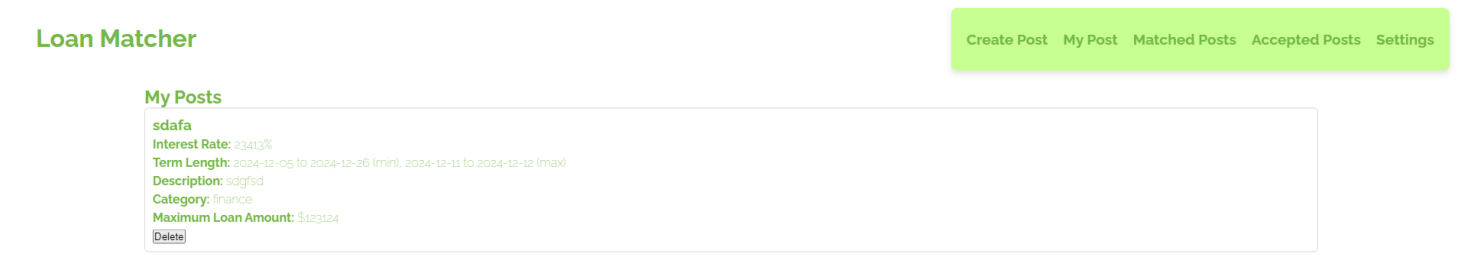

Client/LP settings tab toggled:



Client/LP logout confirmation:




Client/LP My Post tab populated:



**Appendix:**

Sample Client user:

| CEmail | Fname | Lname | Credit | CPassword | CLocation | DOB_Establishment | CPhone |
|---|---|---|---|---|---|---|---|
| johndoe@gmail.com | John | Doe | 666 | 11111111 | Johnistan | 2024-12-07 | 1234567890 |

Sample Loan provider user:

| LEmail | Fname | Lname | LPassword | LLocation | LPhone | LicenseNo |
|---|---|---|---|---|---|---|
| janedoe@gmail.com | Jane | Doe | 11111111 | Johnistan | 0987654321 | 123456 |

Sample Loan Posts:

| P_ID | PTitle | Description | MaxRate | MinTermStart | MinTermEnd | MaxTermStart | MaxTermEnd | PStatus | LType | Desired_amt |
|---|---|---|---|---|---|---|---|---|---|---|
| 9 | Loan Offer A | Low interest rate loan for small businesses. | 5 | 2024-01-01 | 2025-01-01 | 2024-12-01 | 2026-12-31 | Pending | business | 10000 |
| 10 | Loan Offer B | Flexible term lan for real estate projects. | 3 | 2024-02-15 | 2025-02-15 | 2024-02-15 | 2027-02-15 | Pending | real-estate | 50000 |
| 11 | Loan Offer C | Short-term loan for startups. | 4 | 2024-03-01 | 2025-03-01 | 2024-03-01 | 2026-03-01 | Pending | other | 20000 |
| 12 | Loan Offer D | Educational loan with no collateral | 4 | 2024-05-01 | 2025-02-01 | 2024-05-01 | 2026-05-01 | Pending | other | 15000 |

Sample Loan Offers:

| OName | MinRate | MinTermStart | MinTermEnd | MaxTermStart | MaxTermEnd | OStatus | LType | O_ID | Min_amt | Max_amt | Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Loan Offer A | 5 | 2024-01-01 | 2025-01-01 | 2024-01-01 | 2026-12-31 | Pending | business | 2 | 10000 | 20000 | Low interest rate loan for small businesses |
| Loan Offer B | 3 | 2024-02-15 | 2025-02-15 | 2024-02-15 | 2027-02-15 | Pending | real-estate | 3 | 50000 | 70000 | Flexible term loan for real estate projects. |
| Loan Offer C | 4 | 2024-03-01 | 2026-03-01 | 2024-03-01 | 2026-03-01 | Pending | other | 4 | 20000 | 50000 | Short-term loan for startups. |
| Loan Offer D | 4 | 2024-05-01 | 2025-05-01 | 2024-05-01 | 2026-03-01 | Pending | other | 5 | 15000 | 30000 | Educational loan with no collateral. |